

Measured Geodesic Laminations (Bradley Zykoski)

What is a lamination?

- Let S be a ~~surface~~ ^{of} surface of finite type. Succinctly, a lamination of S is a foliation of some closed subset of S . Let us assume S admits a hyperbolic metric.
- A geodesic lamination is a lamination all of whose leaves are geodesics (necessarily simple, though maybe not all closed, and all disjointly embedded).
- An easy example of a geodesic lamination is just a finite collection of disjoint simple closed geodesics.
- Using e.g. interval exchange maps (cf. Bonahon), one can construct geodesic laminations with uncountably many leaves, all of which are infinite (i.e. parametrized by $\mathbb{R}$ rather than S^1).
- For the same reason as with foliations, we would like a notion of geometric intersection number (recall that for a foliation with measure μ , the intersection with a homotopy class α of curves is calculated as $\inf_{\alpha_0 \sim \alpha} \int d\mu$). We will similarly associate a measure to our geodesic laminations.
- Indeed, this analogy with foliations is rather direct. One can construct a homeomorphism between the space of measured foliations on S and the space of measured laminations on S , where the map takes a foliation

to the lamination given by straightening the foliation's leaves into geodesics.
(cf. Leininger)

Measured geodesic laminations

Defn: A transverse measure on a geodesic lamination λ is a collection of (Radon) measures $\{\mu_\alpha \mid \alpha \text{ an arc on } S \text{ transverse to the leaves of } \lambda\}$, where each μ_α is a measure on α satisfying:

1. If α' is a subarc of α , then $\mu_{\alpha'}$ is the restriction of μ_α to α' .
2. If α is homotopic to α' through arcs transverse to λ via a homotopy h_t , then $\mu_{\alpha'}$ is the induced measure $(h_1)_*(\mu_\alpha)$.
3. The support of μ_α is precisely $\alpha \cap \lambda$.

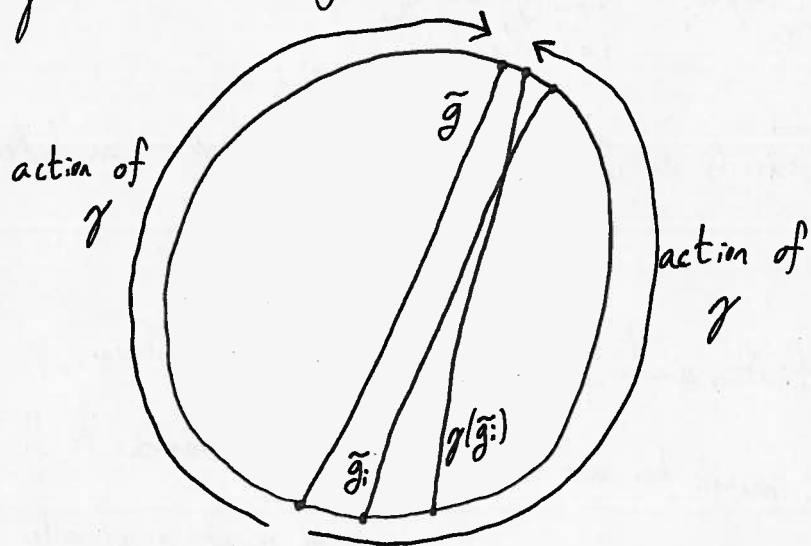
• To exemplify the key features of transverse measures, we give a proof of the following lemma.

• Lemma (Prop. 1.7.2 of Penner/Harer): Suppose g is a leaf of a ^(compactly supported) measured geodesic lamination λ . If g is simple closed, then g is isolated from $\lambda - g$. If g is an infinite geodesic, then every point of every compact subarc $\alpha \subset g$ is a limit point of $g - \alpha$.

• Proof: If g is simple closed and not isolated, then let $\tilde{g}$ be a lift of g to the universal cover of S , and let $\gamma \in \pi_1(S) \subset \text{PSL}(2, \mathbb{R})$ be a hyperbolic transformation with axis $\tilde{g}$. We will show there is some leaf g' of λ with a lift $\tilde{g}'$ asymptotically parallel to $\tilde{g}$. Otherwise, since g is not isolated, there is a sequence of leaves g_i of λ with lifts $\tilde{g}_i$ whose endpoints on $\partial \mathbb{H}^2$

converge to those of $\tilde{g}$. (The endpoints must do this, or else this countable family could not approximate g everywhere).

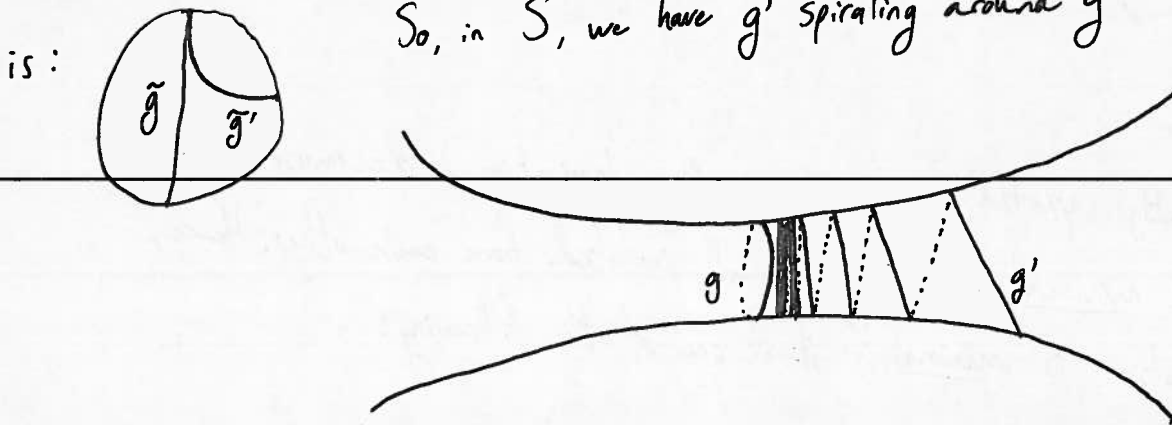
The leaves of λ are disjointly embedded, and so $\tilde{g}$ is disjoint from the $\tilde{g}_i$. By our assumption, none of the $\tilde{g}_i$ are asymptotically parallel to $\tilde{g}$. Consider the action of γ on the $\tilde{g}_i$. For some great enough i , the endpoints of $\tilde{g}_i$ will be close enough to those of $\tilde{g}$ so that we have the following picture:



But $\gamma(\tilde{g}_i)$ is also a lift of g_i since $\gamma \in \pi_1(S)$, and this contradicts our assumption that the leaves of λ are disjointly embedded. Hence there is indeed some leaf $g' \in \lambda$ with a lift $\tilde{g}'$ asymptotically parallel to $\tilde{g}$.

So far, we haven't used our transverse measure at all! Our lifted picture

So, in S , we have g' spiraling around g :



Consider an arc α in S transverse to λ and intersecting both g and g' . The arc α must intersect g' at infinitely many points $\{x_i\}$. Let α_i be subarcs of α about the x_i , small enough that $\alpha_i \cap \lambda = x_i$ for each i . Note that the α_i are homotopic to each other through arcs transverse to λ : just slide them along g' ! But then we have

$$\int_{\alpha} d\mu_{\alpha} \geq \sum_{i=1}^{\infty} \int_{\alpha_i} d\mu_{\alpha_i} = \sum_{i=1}^{\infty} \int_{\alpha_1} d\mu_{\alpha_1} = \infty$$

because $\int_{\alpha_1} d\mu_{\alpha_1} > 0$. This is a contradiction, and so we conclude that g is isolated.

Now suppose g is infinite, and $p \in \alpha \subset g$, α a compact subarc, p isolated from $g \setminus \alpha$. Then there must be an arc β in S transverse to λ meeting g only at p . Therefore any transverse β' must meet g at only finitely many points; otherwise $\int_{\beta'} d\mu_{\beta'} \geq \sum_{i=1}^{\infty} \int_{\beta} d\mu_{\beta} = \infty$ as before. Let $\{\alpha_j\}$ be a pants decomposition of S_0 with each α_j transverse to λ . Then g must meet each α_j finitely many times, contradicting our assumption that g is infinite. $\square$

Corollary: A (compactly supported) measured geodesic lamination λ decomposes into finitely many closed minimal laminations, all isolated from each other. Here, we say a lamination is minimal if it is one of the following:

1. A closed geodesic with atomic transverse measure.

2. A measured geodesic lamination λ' such that every leaf of λ' is infinite and dense in λ . □

Furthermore, none of these take up any area on the surface S .

• Claim: Let λ be a geodesic lamination on S . Then $S - \lambda$ has finitely many components, and λ has zero area.

• Proof sketch: By Gauss-Bonnet, there is a lower bound of π for the area (cf. Thurston) of each connected component of $S - \lambda$. We can extend our lamination to a singular foliation, and then double $S - \lambda$ across its boundary components to get surfaces without boundary. Using the indices of the singularities to compute Euler characteristic, we obtain $2\chi(S) = \chi(\text{double } S - \lambda)$, and hence by Gauss-Bonnet:

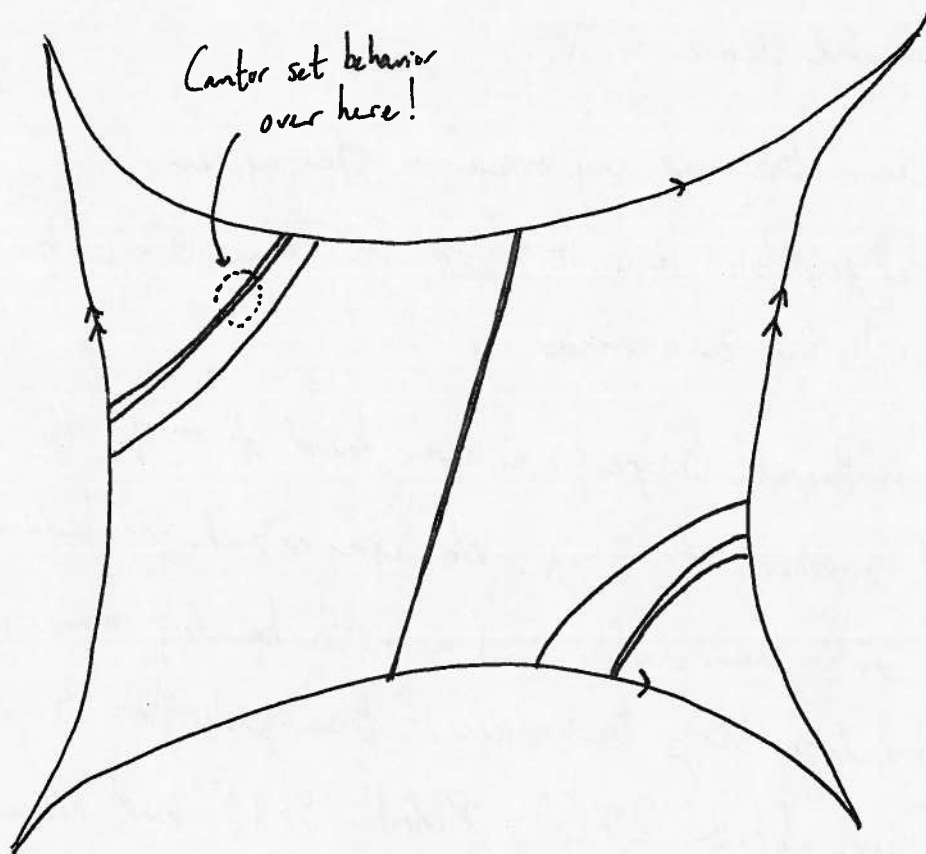
$$\text{Area}(S) = \text{Area}(S - \lambda),$$

Therefore $\text{Area}(\lambda) = 0$.

• By our characterization of measured geodesic laminations λ as unions of finitely many minimal laminations, and the above claim, it follows that any transverse arc intersects λ at the union of finitely many isolated points (#1) and Cantor sets (i.e. closed, perfect, no interior) (#2). □

• This fact can also be derived from a theorem of Birman and Series:

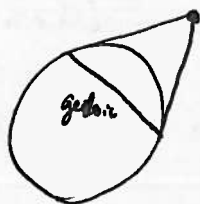
• Theorem (Birman/Series): The union of all simple geodesics on a hyperbolic surface has Hausdorff dimension 1.



An example by Bonahon of a geodesic lamination on the punctured torus

The space of measured laminations

- There is a way of thinking of measured laminations that illustrates that the space of measured laminations does not depend on any particular hyperbolic metric on S .
- The set $G(\tilde{S})$ of geodesics on $\tilde{S}$ can be topologized as a Möbius band in the following way: Each geodesic is determined by a pair of points on $\partial_\infty \tilde{S}$, and this pair is determined by a point of $\mathbb{RP}^2$ lying outside the disk $\tilde{S}$:



(Points at infinity in $\mathbb{RP}^2$ determine antipodal points on $\partial_\infty \tilde{S}$)

- We need not refer to any metric on S in this construction, because the ~~identity~~ identity map $\tilde{S} \rightarrow \tilde{S}$ extends to a homeomorphism $\overline{\tilde{S}} \cong \overline{\tilde{S}}$.
- Now, we have a 1-1 correspondence:

$$\left\{ \begin{array}{l} \text{measured laminations} \\ \text{on } S \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \pi_1(S)\text{-invariant Radon} \\ \text{measures on } G(\tilde{S}) \end{array} \right\}$$

- We indicate one direction of this correspondence:

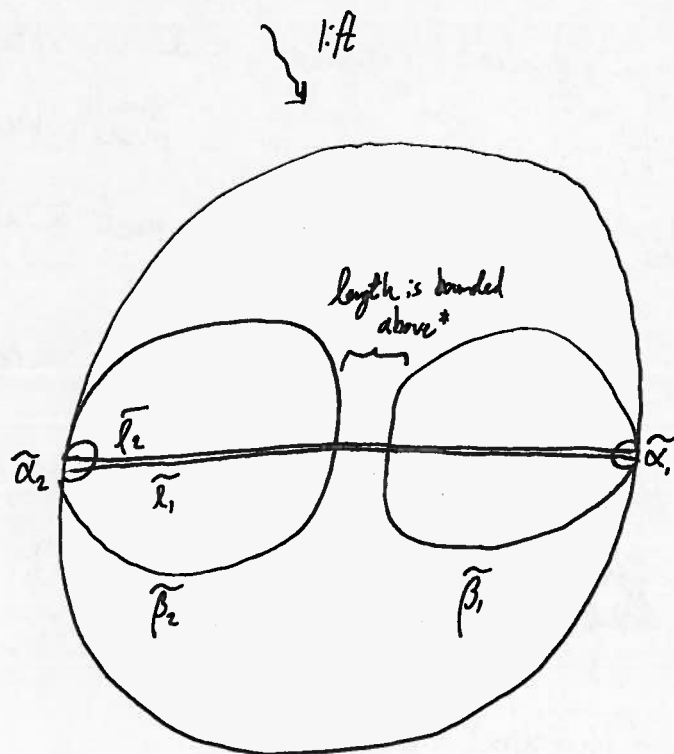
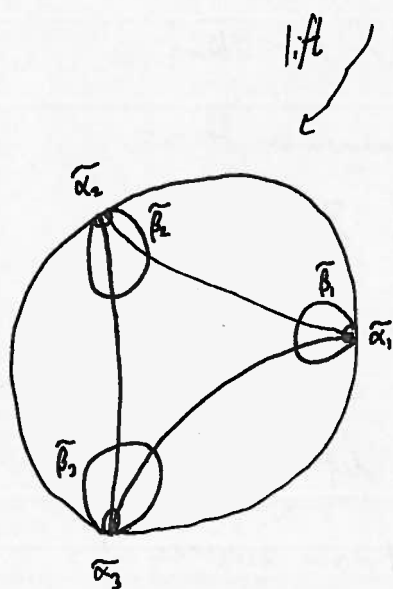
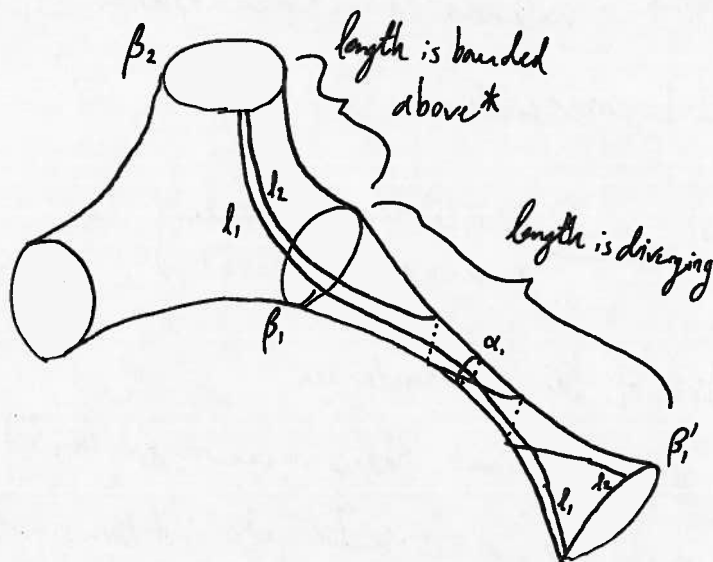
Let μ be a $\pi_1(S)$ -invariant Radon measure on $G(\tilde{S})$, and let α be an arc in S transverse to $\lambda = \text{projection onto } S \text{ of the support of } \mu$. Then λ is a geodesic lamination, and we obtain a transverse measure as follows. Lift α to $\tilde{S}$, and for a ^{small} measurable subset $A \subset \tilde{\alpha}$ let $\tilde{A} \subset G(\tilde{S})$ be the collection of geodesics that meet $\tilde{\alpha}$ at A . Then define $\int_A d\mu_\alpha := \int_{\tilde{A}} d\mu$.

- This all goes to show that the space $MI(S)$ can be defined independent of the hyperbolic metric on S . The space $MI(S)$ is topologized as a subspace of the ~~vector~~ ^{vector} space of measures on $G(\tilde{S})$.

Dehn-Thurston coordinates:

- Lemma: We can always choose a pants decomposition $\{\alpha_i\}$ of S with short enough α_i that if we take non-geodesic pants P_j bounded by curves β_i, β_i' homotopic to α_i of length $1/2$, then any two simple geodesics l_1, l_2 passing through P_j through the same cuffs remain within some fixed ε of each other while inside the pants.

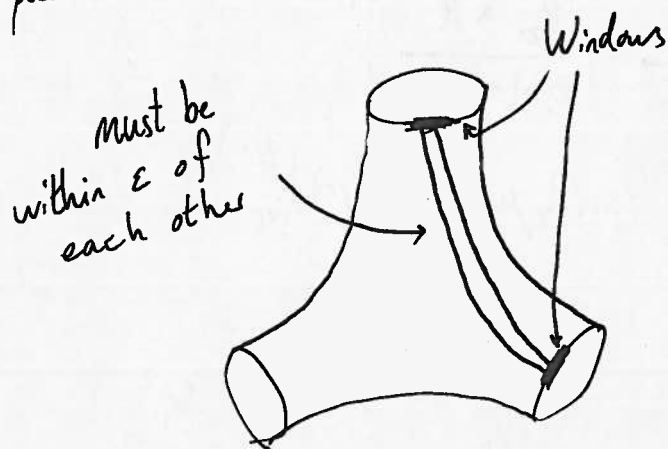
• Proof: As $\text{length}(\alpha_i) \rightarrow 0 \dots$



* As $\text{length}(\alpha_i) \rightarrow 0$, the right-angled hexagon tends to an ideal triangle, giving cusps on the limiting surface. Curves about cusps have $\text{length} \rightarrow 0$ as they tighten around the cusp, and since our β_i 's all have length $1/2$, it therefore follows that on the limiting surface, the β_i 's have bounded distance from each other. Hence there is an upper bound on this distance between the β_i 's as $\text{length}(\alpha_i) \rightarrow 0$.

II

It follows from the lemma that we can choose α_i 's short enough that as any simple geodesic goes through P_j , it enters and ~~exit~~ exits through small intervals ("windows") on the β_i , and hence we know exactly what it does inside the pants:



Therefore we can measure all of the interesting behavior of ~~the~~ a geodesic by looking at what happens inside the collars C_i about the α_i that are bounded by the $1/2$ -length β_i, β_i' .

We now define the Dehn-Thurston coordinates on $ML(S)$. For each ~~of~~ collar C_i , let γ_i be a geodesic arc from one boundary circle β_i of C_i to the other, β_i' , such that γ_i does not meet the windows. Any geodesic lamination must intersect γ_i transversely, so we may define a twist coordinate:

$$t_i(\lambda, \mu) = \int_{\gamma_i} d\mu$$

If none of the α_i are leaves of λ , then the orientation of our ~~the~~ surface allows us to give this twist coordinate a well-defined sign determined by the handedness of the intersection of λ with γ_i . Further, we have the intersection number of λ with α_i :

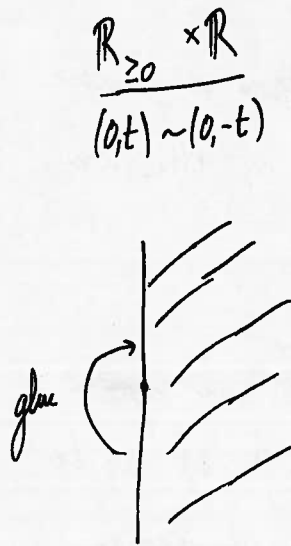
$$s_i(\lambda, \mu) = \int_{\alpha_i} d\mu.$$

If α_i is a leaf of λ , then there is no notion of handedness for the intersection of λ with γ_i . We must resign ourselves to choosing both signs, i.e. projectivizing the i^{th} twist coordinate when $\alpha_i \subset \lambda$. Hence we have a map:

$$\Phi: \mathcal{ML}(S) \rightarrow \left(\frac{\mathbb{R}_{\geq 0} \times \mathbb{R}}{(0,t) \sim (0,-t)} \right)^{3g-3}$$

$$(\lambda, \mu) \mapsto (s_i(\lambda, \mu), t_i(\lambda, \mu))_{i=1}^{3g-3}$$

But:



$\cong$



and so $\Phi: \mathcal{ML}(S) \rightarrow (\mathbb{R}^2)^{3g-3} = \mathbb{R}^{6g-6}$.

Theorem: The map Φ is a homeomorphism onto $\left(\frac{\mathbb{R}_{\geq 0} \times \mathbb{R}}{(0,t) \sim (0,-t)} \right)^{3g-3} \setminus \{0\}$,

and ~~and~~ respects scaling by positive real numbers.

Corollary: The subset of $\mathcal{ML}(S)$ consisting of (λ, μ) where λ is a finite union of simple closed geodesics is dense in $\mathcal{ML}(S)$.

• Proof: If $\Phi(\lambda, \mu) \in \left(\frac{\mathbb{R}_{20} \times \mathbb{R}}{(0, t) \sim (0, -t)} \right)^{3g-3}$ has integral coordinates, then λ is a finite union of simple closed geodesics. The set of positive real multiples of integral points in $\left(\frac{\mathbb{R}_{20} \times \mathbb{R}}{(0, t) \sim (0, -t)} \right)^{3g-3}$ is dense, and hence its preimage under Φ is dense in $MS(S)$. $\square$

• Why do we care?

- Measured geodesic laminations are an important ingredient in many constructions, e.g. Thurston's compactification of Teichmüller space, Thurston's earthquake maps, and pleated surfaces.

• References

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